

Slide 9 Proof

1 Recapitulation

We introduced the method of maximum likelihood for simple linear regression in the notes for two lectures ago. Let's review.

We start with the statistical model, which is the Gaussian-noise simple linear regression model, defined as follows:

1. The distribution of X is arbitrary (and perhaps X is even non-random).
2. If $X = x$, then $Y = \beta_0 + \beta_1 x + \epsilon$, for some constants ("coefficients", "parameters") β_0 and β_1 , and some random noise variable ϵ .
3. $\epsilon \sim N(0, \sigma^2)$, and is independent of X .
4. ϵ is independent across observations.

A consequence of these assumptions is that the response variable Y is independent across observations, conditional on the predictor X , i.e., Y_1 and Y_2 are independent given X_1 and X_2 (Exercise 1).

As you'll recall, this is a special case of the simple linear regression model: the first two assumptions are the same, but we are now assuming much more about the noise variable ϵ : it's not just mean zero with constant variance, but it has a particular distribution (Gaussian), and everything we said was uncorrelated before we now strengthen to independence¹.

Because of these stronger assumptions, the model tells us the conditional pdf of Y for each x , $p(y|X = x; \beta_0, \beta_1, \sigma^2)$. (This notation separates the random variables from the parameters.) Given any data set $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, we can now write down the probability density, under the model, of seeing that data:

$$\prod_{i=1}^n p(y_i|x_i; \beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (\beta_0 + \beta_1 x_i))^2}{2\sigma^2}}$$

¹See the notes for lecture 1 for a reminder, with an explicit example, of how uncorrelated random variables can nonetheless be strongly statistically dependent.

In multiplying together the probabilities like this, we are using the independence of the Y_i .

When we see the data, we do not *known* the true parameters, but any guess at them, say (b_0, b_1, s^2) , gives us a probability density:

$$\prod_{i=1}^n p(y_i|x_i; b_0, b_1, s^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi s^2}} e^{-\frac{(y_i - (b_0 + b_1 x_i))^2}{2s^2}}$$

This is the **likelihood**, a function of the parameter values. It's just as informative, and much more convenient, to work with the **log-likelihood**,

$$L(b_0, b_1, s^2) = \log \prod_{i=1}^n p(y_i|x_i; b_0, b_1, s^2) \quad (1)$$

$$= \sum_{i=1}^n \log p(y_i|x_i; b_0, b_1, s^2) \quad (2)$$

$$= -\frac{n}{2} \log 2\pi - n \log s - \frac{1}{2s^2} \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))^2 \quad (3)$$

In the **method of maximum likelihood**, we pick the parameter values which maximize the likelihood, or, equivalently, maximize the log-likelihood. After some calculus (see notes for lecture 5), this gives us the following estimators:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{c_{XY}}{s_X^2} \quad (4)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \quad (5)$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2 \quad (6)$$

As you will recall, the estimators for the slope and the intercept exactly match the least squares estimators. This is a special property of assuming independent Gaussian noise. Similarly, $\hat{\sigma}^2$ is exactly the in-sample mean squared error.

Slide 10 Proof

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \text{Var}\left(\frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}\right) \\&= \text{Var}\left(\frac{\sum_i (x_i - \bar{x})(\beta_0 + \beta_1 x_i + u_i)}{\sum_i (x_i - \bar{x})^2}\right), \quad \text{substituting in the above} \\&= \text{Var}\left(\frac{\sum_i (x_i - \bar{x})u_i}{\sum_i (x_i - \bar{x})^2}\right), \quad \text{noting only } u_i \text{ is a random variable} \\&= \frac{\sum_i (x_i - \bar{x})^2 \text{Var}(u_i)}{(\sum_i (x_i - \bar{x})^2)^2}, \quad \text{independence of } u_i \text{ and, } \text{Var}(kX) = k^2 \text{Var}(X) \\&= \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}\end{aligned}$$